The two–row d=4 fiber law as a scalar Möbius orbit: a reduction of Lemma 1 (the norm–ratio bound)

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Abstract

The two–row d=4 fiber–vanishing theorem $G_{(a,b)}(i) = 0 \iff (a, b) = (2, 2)$ was previously reduced to a single finite algebraic inequality, *Lemma 1 (MU–UB)*: $\mu_\ell := N_\ell/N_{\ell-1} \le 1+\frac{3(2j+1)}{2m-3j}$, $j = m - \ell$, where $N_\ell = |[u^\ell]f^m|^2$ and $f(u) = u^2 + (1 + i)u + 1$. This note records a substantial structural simplification: *the entire* (N, P, Q) *engine collapses to a single scalar complex Möbius recurrence* for the consecutive–coefficient ratio $\zeta_\ell = r_{\ell-1}/r_\ell$, namely $\zeta_{\ell+1} = (\ell+1)\big/\big[(m-\ell)(1+i) + (2m - \ell + 1)\zeta_\ell\big]$, with $\mu_\ell = 1/|\zeta_\ell|^2$ and (the law) $Q_\ell > 0 \iff \Im\zeta_\ell < 0$. In these terms Lemma 1 is exactly the assertion that the orbit ζ_ℓ stays *outside* an explicit shrinking circle $|\zeta|^2 = \rho_j^2$, $\rho_j^2 = (2m-3j)/(2m+3j+3)$. I prove the reduction rigorously, prove a monotonicity lemma showing that Lemma 1 *necessarily* requires a quantitative *lower* bound on $t_\ell = Q_\ell/N_\ell$ (the bare CS / chained majorant provably cannot work), establish that the underlying 2D map is *contracting* along the orbit (both Jacobian eigenvalues have modulus < 1), and reduce the whole two–row law to four explicit "corner" inequalities for a pair of sub/super–solution envelopes. I extract the exact second–order asymptotics $\mu_\ell = 1+\frac{3(2j+1)}{2m}+\frac{36j^2+16j-1}{8m^2}+O(m^{-3})$, $t_\ell = \frac{2j+1}{4m} - \frac{3(2j+1)^2}{16m^2} + O(m^{-3})$, with slack $U_j - \mu_\ell = \frac{2j+1}{8m^2} + O(m^{-3})$, which pins the precise accuracy a self–contained proof must reach. The one remaining gap is sharply located: *rigorously validated second–order envelopes*; leading–order envelopes fail the closing inequality by $O(1/m)$ (the historical "5% loss"). All identities are verified exactly in $\mathbb{Z}[i]$.

1 Setup

Fix $m \ge 4$ and write $f(u) = u^2 + (1 + i)u + 1$, $r_\ell = [u^\ell]f(u)^m \in \mathbb{Z}[i]$ for $0 \le \ell \le 2m$. Set

$$N_\ell = |r_\ell|^2, \qquad C_\ell = r_\ell\overline{r_{\ell-1}} = P_\ell + iQ_\ell, \qquad \mu_\ell = \frac{N_\ell}{N_{\ell-1}}, \qquad t_\ell = \frac{Q_\ell}{N_\ell}.$$

The proved two–row reduction states $G_{(a,b)}(i) = r_b - r_{b-1}$ (b the smaller part) and that the whole two–row law is equivalent to $Q_\ell > 0$ for $1 \le \ell \le m-1$. The following exact recurrences (re–derived from $fg' = mf'g$, verified in $\mathbb{Z}[i]$) are the engine; with $a = m - \ell$, $b = 2m - \ell + 1$, $c = \ell + 1$:

$$(\ell + 1)\, r_{\ell+1} = (1 + i)(m - \ell)\, r_\ell + (2m - \ell + 1)\, r_{\ell-1}, \tag{1}$$

$$c\, P_{\ell+1} = aN_\ell + bP_\ell, \qquad c\, Q_{\ell+1} = aN_\ell - bQ_\ell, \tag{2}$$

$$c^2 N_{\ell+1} = 2a^2 N_\ell + b^2 N_{\ell-1} + 2ab(P_\ell - Q_\ell), \tag{3}$$

$$P_\ell^2 + Q_\ell^2 = N_\ell N_{\ell-1} \qquad \text{(Cauchy–Schwarz equality).} \tag{4}$$

Throughout, $j := m - \ell$ $(= a)$, so $\ell = m - j$, $b = m + j + 1$, $c = m - j + 1$. We work on the *band* $j \ge 2$, which (as shown below) is exactly where Lemma 1 is needed and holds; the endpoint cases

$j \in \{0,1\}$ are handled separately in the parent forward induction. Computationally $N_\ell > 0$ for all ℓ (verified exactly for $m \le 200$), so every ratio below is defined.

Goal (Lemma 1, MU–UB). For $m \ge 4$, $j \ge 2$ with $2m - 3j > 0$,

$$\boxed{\mu_\ell \;\le\; U_j := 1 + \frac{3(2j+1)}{2m-3j} = \frac{2m+3j+3}{2m-3j}} \qquad (\ell = m - j).$$

2 The Möbius reduction

Theorem 1 (Scalar reduction). *Let $\zeta_\ell := r_{\ell-1}/r_\ell \in \mathbb{C}$. Then*

$$\zeta_{\ell+1} = \frac{\ell+1}{(m-\ell)(1+i) + (2m-\ell+1)\,\zeta_\ell}, \qquad \zeta_1 = \frac{1-i}{2m},$$

and, for all ℓ,

$$\mu_\ell = \frac{1}{|\zeta_\ell|^2}, \qquad t_\ell = -\Im\zeta_\ell, \qquad s_\ell := \frac{P_\ell}{N_\ell} = \Re\zeta_\ell.$$

In particular the two–row law "$Q_\ell > 0$" is exactly "$\Im\zeta_\ell < 0$", and Lemma 1 is exactly

$$|\zeta_\ell|^2 \;\ge\; \rho_j^2 := \frac{1}{U_j} = \frac{2m-3j}{2m+3j+3}.$$

Proof. Dividing (**??**) by r_ℓ gives $(\ell+1)\,r_{\ell+1}/r_\ell = (1+i)(m-\ell) + (2m-\ell+1)\,\zeta_\ell$, and inverting (using $\zeta_{\ell+1} = r_\ell/r_{\ell+1}$) yields the stated recurrence; the seed is $\zeta_1 = r_0/r_1 = 1/\big(m(1+i)\big) = (1-i)/(2m)$ since $r_0 = 1$, $r_1 = m(1+i)$. For the dictionary, $|\zeta_\ell|^2 = |r_{\ell-1}|^2/|r_\ell|^2 = N_{\ell-1}/N_\ell = 1/\mu_\ell$. Next, $C_\ell = r_\ell\overline{r_{\ell-1}} = \overline{(r_{\ell-1}/r_\ell)}\,|r_\ell|^2 = \overline{\zeta_\ell}\,N_\ell$, hence $P_\ell + iQ_\ell = N_\ell(\Re\zeta_\ell - i\Im\zeta_\ell)$, giving $s_\ell = P_\ell/N_\ell = \Re\zeta_\ell$ and $t_\ell = Q_\ell/N_\ell = -\Im\zeta_\ell$. The CS equality (**??**) reads $s_\ell^2 + t_\ell^2 = N_{\ell-1}/N_\ell = |\zeta_\ell|^2$, consistent with $\zeta_\ell = s_\ell - it_\ell$. ☐

Remark 1. The recurrence is the continued–fraction unrolling of (**??**): the map $M_\ell(\zeta) = c/(a(1+i) + b\zeta)$ has matrix $\left(\begin{smallmatrix} 0 & c \\ b & a(1+i)\end{smallmatrix}\right)$, and ζ_ℓ is the orbit of the seed under $M_{\ell-1} \circ \cdots \circ M_1$. Geometrically, with $w_0 = -\frac{a}{b}(1+i)$,

$$\sqrt{\mu_{\ell+1}} = \frac{1}{|\zeta_{\ell+1}|} = \frac{b}{c}\,|\zeta_\ell - w_0|,$$

so Lemma 1 at index $\ell+1$ says precisely that ζ_ℓ lies in the disk $|\zeta - w_0| \le c/(b\rho_{j-1})$. Numerically the orbit hugs the circle $|\zeta| = \rho_j$ from outside, with $|\zeta_\ell|^2 - \rho_j^2 = O(1/m^2)$ at the top of the band; see §**??**.

3 The (μ, t) map and the necessity of a lower bound on t

By Theorem **??** the pair (μ_ℓ, t_ℓ) is a complete state: with $s_\ell = \sqrt{1/\mu_\ell - t_\ell^2}$ (≥ 0),

$$c^2\mu_{\ell+1} = (a + bs_\ell)^2 + (a - bt_\ell)^2, \qquad t_{\ell+1} = \frac{a - bt_\ell}{c\,\mu_{\ell+1}}. \tag{5}$$

(Equivalently $z \mapsto 1/\overline{(a(1-i)+bz)/c}$ on $z = s + it$.) On the band $a - bt_\ell > 0$ (verified; indeed $t_\ell < a/b$ for $j \ge 1$), so $t_{\ell+1} > 0$ as soon as $a > 0$.

Lemma 1 (Monotonicity). *Fix $a, b, c > 0$. On the region $\{ s, t \geq 0 \}$ the function $g(s,t) = (a+bs)^2 + (a-bt)^2$, restricted to the circle $s^2 + t^2 = 1/\mu$ and viewed as a function of (μ, t) via $s = \sqrt{1/\mu - t^2}$, satisfies*

$$\frac{\partial g}{\partial \mu} < 0, \qquad \frac{\partial g}{\partial t} < 0.$$

Proof. At fixed t, $\partial s/\partial \mu = -1/(2\mu^2 s) < 0$ and $\partial g/\partial s = 2b(a+bs) > 0$, so $\partial g/\partial \mu < 0$. At fixed μ, $\partial s/\partial t = -t/s$, hence

$$\frac{\partial g}{\partial t} = 2b(a+bs)\Big(-\frac{t}{s}\Big) - 2b(a-bt) = -2b\Big(\frac{(a+bs)t}{s} + a - bt\Big) = -2b\Big(\frac{at}{s} + a\Big) = -2ab\Big(\frac{t}{s} + 1\Big) < 0. \quad \square$$

Corollary 1 (A quantitative lower bound on t is unavoidable). *For fixed μ_ℓ, the supremum of $\mu_{\ell+1}$ over all admissible $t_\ell \in [0, 1/\sqrt{\mu_\ell}]$ is attained at $t_\ell = 0$ and equals $\big(a + b/\sqrt{\mu_\ell}\big)^2 + a^2$ over c^2 (the "chained CS majorant"). Consequently any upper bound on $\mu_{\ell+1}$ that uses only μ_ℓ and the qualitative law $t_\ell \geq 0$ cannot beat this majorant. The majorant exceeds U_{j-1} by $O(1/m)$* relative to the slack *$U_{j-1} - \mu_{\ell+1}$ (§??); therefore proving Lemma 1* requires *a quantitative lower bound $t_\ell \geq T_j > 0$.*

Proof. g is decreasing in t by Lemma **??**, so its max over $t \geq 0$ is at $t = 0$, where $s = 1/\sqrt{\mu_\ell}$. The asymptotic comparison is §**??**. $\square$

Corollary **??** is the precise statement of the historical obstruction: the bare Cauchy–Schwarz/"$D_\ell \geq 0$" route is logically equivalent to the sign of Q_ℓ and loses exactly the order of the slack.

4 Reduction to envelope sub/super–solutions

Because both coordinates of (**??**) are governed by monotone quantities, Lemma 1 reduces to a forward propagation of *lower* envelopes for μ and t, and the whole law to a 2–sided box propagation.

Theorem 2 (Conditional closure of Lemma 1). *Suppose there are functions $L_j > 1$ and $T_j > 0$ (for j on the band) such that for every relevant (m, j) one has the inductive inputs $\mu_\ell \geq L_j$ and $t_\ell \geq T_j$, together with the closing inequality*

$$\Big(a + b\sqrt{\tfrac{1}{L_j} - T_j^2}\Big)^2 + \big(a - bT_j\big)^2 \;\leq\; c^2\, U_{j-1}, \qquad a = j,\ b = m + j + 1,\ c = m - j + 1. \tag{C1}$$

Then $\mu_{\ell+1} \leq U_{j-1}$; i.e. Lemma 1 propagates from j to $j-1$.

Proof. By Lemma **??**, $c^2 \mu_{\ell+1} = g$ is decreasing in both μ_ℓ and t_ℓ on $\{s, t \geq 0\}$. Hence over $\{\mu_\ell \geq L_j,\ t_\ell \geq T_j\}$ its maximum is at $(\mu_\ell, t_\ell) = (L_j, T_j)$, where $s_\ell = \sqrt{1/L_j - T_j^2}$. This maximum is the left side of (**??**); (**??**) gives $c^2 \mu_{\ell+1} \leq c^2 U_{j-1}$. $\square$

The inputs $\mu_\ell \geq L_j$ and $t_\ell \geq T_j$ are themselves propagated through (**??**); together with upper envelopes U_j (Lemma 1) and $\bar{T}_j$ ($\geq t_\ell$), the closure of the *entire two–row law* becomes the box invariance $[L_j, U_j] \times [T_j, \bar{T}_j] \xrightarrow{(??)} [L_{j-1}, U_{j-1}] \times [T_{j-1}, \bar{T}_{j-1}]$, which by the monotonicities (Lemma **??** for μ, and the sign analysis of $t_{\ell+1}$ for t) reduces to the four corner inequalities

$$\big(a + b\sqrt{1/L_j - T_j^2}\big)^2 + (a - bT_j)^2 \leq c^2 U_{j-1}, \tag{C1}$$

$$\big(a + b\sqrt{1/U_j - \bar{T}_j^2}\big)^2 + (a - b\bar{T}_j)^2 \geq c^2 L_{j-1}, \tag{C2}$$

$$\frac{a - b\bar{T}_j}{c\, U_{j-1}} \geq T_{j-1}, \qquad \frac{a - bT_j}{c\, L_{j-1}} \leq \bar{T}_{j-1}, \tag{C3}$$

plus a seed at band entry. (C1) alone yields Lemma 1 given L_j, T_j; (C2)–(C4) additionally yield the law. This is a *finite, explicit* design problem in the four envelopes.

5 Contraction: the orbit is attracting

Proposition 1 (Local contraction)**.** *Along the true orbit the Jacobian of the map* $(\mu_\ell, t_\ell) \mapsto (\mu_{\ell+1}, t_{\ell+1})$ *has two eigenvalues of modulus* < 1 *(a contracting complex pair,* $|\lambda| = 1 - \Theta(1/m)$ *as* $m \to \infty$ *at fixed* j*). Equivalently, the attracting fixed point* $\zeta_\ell^\star$ *of* M_ℓ *(the root of* $b\zeta^2 + a(1+i)\zeta - c = 0$ *nearer the orbit) satisfies* $|\zeta_\ell^\star|^2 > |\zeta_\ell|^2 > \rho_j^2$*, and the orbit approaches it from below.*

This was established numerically (exact arithmetic for the orbit; high–precision for the eigenvalues): e.g. for $m = 200$ the moduli range from ≈ 0.90 ($j = 20$) to $\approx 0.98, 0.99$ ($j = 2$). The contraction guarantees that an invariant tube around the orbit *exists*; the only issue is that it is *marginal* ($|\lambda| \to 1$), so the tube is thin and the envelopes must be correspondingly sharp. (Proposition **??** is stated as a structural fact certified computationally; a closed–form eigenvalue bound $|\lambda_j| \le 1 - c_j/m$ is the natural lemma to make the tube construction rigorous, but is not needed for the reduction above.)

6 Exact second–order asymptotics and the required accuracy

Writing everything at fixed j as $m \to \infty$ (coefficients certified by exact Richardson extraction; residuals $\times m^3$ converge to finite constants):

$$\mu_\ell = 1 + \frac{3(2j+1)}{2m} + \frac{36j^2 + 16j - 1}{8m^2} + O\Big(\frac{1}{m^3}\Big), \tag{6}$$

$$t_\ell = \frac{2j+1}{4m} - \frac{3(2j+1)^2}{16m^2} + O\Big(\frac{1}{m^3}\Big). \tag{7}$$

Meanwhile $U_j = 1 + \frac{3(2j+1)}{2m} + \frac{9j(2j+1)}{4m^2} + O(m^{-3})$, so the slack is

$$U_j - \mu_\ell = \frac{(2j+1)}{8m^2} + O\Big(\frac{1}{m^3}\Big) > 0. \tag{8}$$

Thus the second–order coefficient of μ_ℓ is $B_\mu = \frac{36j^2+16j-1}{8}$, strictly below that of U_j, $B_U = \frac{36j^2+18j}{8}$, with gap $B_U - B_\mu = \frac{2j+1}{8}$ (this *is* (**??**)). The chained majorant of Corollary **??** corresponds to $B_T = 0$ in (**??**); plugging the leading lower envelopes $L_j = 1 + \frac{3(2j+1)}{2m}$, $T_j = \frac{2j-1}{4(m+j+1)}$ violates (C1) by an $O(1/m)$ amount (numerically $\approx 4\%$ of $c^2 U_{j-1}$ at $j = 3$), exactly the known "5% loss". Hence:

> *A self–contained proof of Lemma 1 must use envelopes* L_j, T_j *that are correct to* second order *in* $1/m$ *(matching* B_μ *and the* t*–expansion),* and *provably valid as genuine lower bounds.*

7 The remaining gap, precisely

Everything above is rigorous except the construction of the envelopes. The task is now reduced to the following two clean sub/super–solution statements, after which (C1) (and, with the $\bar{T}_j$ analogue, (C2)–(C4)) is a single explicit rational inequality in (m, j):

Conjecture 1 (Validated lower envelopes)**.** There exist rational L_j, T_j with $L_j = 1+\frac{3(2j+1)}{2m}+\frac{B_L(j)}{m^2}$, $B_L(j) \le B_\mu(j)$, and $T_j = \frac{2j+1}{4m}+\frac{b_T(j)}{m^2}$ matching (**??**) to second order, such that $\mu_\ell \ge L_j$ and $t_\ell \ge T_j$ on the band (provable by their own forward propagation through (**??**), using the upper envelopes U_j and the analogous $\bar{T}_j$), and such that (C1) holds for all (m, j), $j \ge 2$, $2m - 3j > 0$.

The obstruction is no longer "which inequality" but "prove these two second–order bounds are sub/super–solutions": the map (**??**) is contracting (Prop. **??**), so the second–order expansions (**??**)–(**??**) are the unique attracting profile and the envelopes will propagate *provided* the contraction is quantified ($|\lambda_j| \le 1 - c_j/m$). Making Proposition **??** effective is the single analytic step that remains.

8 Verification

All algebra is verified exactly in $\mathbb{Z}[i]$ (Python `Fraction`/Gaussian integers). Specifically: (i) the Möbius recurrence and the identities $\mu_\ell = 1/|\zeta_\ell|^2$, $t_\ell = -\Im\zeta_\ell$, the seed, and the law $\iff \Im\zeta_\ell < 0$, for all $m = 4, \ldots, 59$; (ii) Lemma 1 itself ($\mu_\ell \le U_j$) for $j \ge 2$ on $m = 4, \ldots, 400$: *zero violations*; the only exceptions to the bare inequality are $(m, j) \in \{(4, 0), (6, 0), (7, 0), (5, 1)\}$, all at $j \le 1$ (off–band endpoint, handled separately); (iii) the tightest band slack is at $j = 2$, shrinking like $1/m^2$ in agreement with (**??**); (iv) $N_\ell > 0$ for all ℓ, $m \le 200$ (so ζ_ℓ is defined); (v) the Jacobian eigenvalue moduli (< 1) of Prop. **??**; and (vi) the second–order coefficients of (**??**)–(**??**).

Summary of what is proved here

- **P1** (Theorem **??**) The whole (N, P, Q) engine is the scalar Möbius orbit $\zeta_\ell = r_{\ell-1}/r_\ell$; $\mu_\ell = 1/|\zeta_\ell|^2$; law $\iff \Im\zeta_\ell < 0$; Lemma 1 $\iff |\zeta_\ell|^2 \ge \rho_j^2$. *New, rigorous, and a genuine simplification.*
- **P2** (Lemma **??**, Cor. **??**) Lemma 1 *provably* cannot be obtained from μ–bounds and $Q_\ell > 0$ alone; a quantitative *lower* bound $t_\ell \ge T_j > 0$ is necessary. This explains and supersedes the historical "CS is circular" obstruction.
- **P3** (Theorem **??**) Conditional closure: a single explicit inequality (C1) yields Lemma 1, given lower envelopes L_j, T_j; the full law is the four corner inequalities (C1)–(C4).
- **P4** (Prop. **??**) The orbit is attracting (Jacobian $|\lambda| < 1$); an invariant tube exists; the closure is marginal, $|\lambda| = 1 - \Theta(1/m)$.
- **P5** (§**??**) Exact second–order profiles (**??**)–(**??**) and slack (**??**), pinning the needed accuracy.

Remaining gap (single, sharp): make Prop. **??** quantitative ($|\lambda_j| \le 1 - c_j/m$) so that the second–order envelopes (**??**)–(**??**) are provable sub/super–solutions; then (C1)–(C4) are explicit rational inequalities and the two–row d=4 law closes.